

Daniella Toth

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EDUCATION

University of Guelph

Guelph, ON

Honours Bachelor of Computing, Software Engineering (Co-op)

Sept. 2023 – April 2028

- **Relevant Coursework:** Software Engineering, Software Design, Database Systems & Concepts, Data Structures & Algorithms, Operating Systems, Software Reliability & Testing, Human-Computer Interaction, Mobile Computing

EXPERIENCE

Front-End Web Developer (Co-op)

August 2025 – December 2025

DBM Systems Inc.

Hybrid, Waterloo, ON

- Rebuilt an Excel-based S&OP planning tool as an interactive web application for supply and demand analysis using Svelte, TypeScript, and CSS.
- Developed a modular data table supporting resizable columns, inline cell editing, multi-cell selection, copy/paste, and full keyboard navigation to match spreadsheet-level usability.
- Partnered directly with a senior developer through code review and design discussions to ship new UI features to production.

Undergraduate Teaching Assistant

Sept. 2024 – Apr. 2026

University of Guelph

Guelph, ON

*CIS*2250: Software Design II*

Jan. 2026 – Apr. 2026

- Supported students in a project-based software design course emphasizing collaboration and industry best practices.
- Ran labs and debugged Python-based team projects, helping students reason through design decisions and trade-offs.

*CIS*1500: Introduction to Programming*

Sept. 2024 – Apr. 2025

- Supported 100+ students through weekly labs, office hours, and discussion forums, debugging code and explaining core programming fundamentals.

PROJECTS

TuneTwin | *Django, Django REST Framework, PostgreSQL, React*

Nov. 2025 – Present

- Co-developed and deployed a full-stack music recommendation app that suggests similar songs from real-time Spotify, Last.fm, and Deezer data.
- Built a Django REST API integrated with three external music APIs, paired with a React frontend for live search and recommendations.

ElementLab | *Vue.js, FastAPI, PostgreSQL, scikit-learn*

Jan. 2026 – May 2026

- Built a full-stack weather analysis platform as Data Engineer & Frontend Developer on an 8-person team, querying 6.4M rows of historical weather data.
- Trained a k-nearest-neighbours model reaching an R^2 of 0.756 for temperature and up to 0.819 for precipitation prediction.

Kerr Black Hole Simulation | *Python, OpenGL/GLSL, GLFW, NumPy*

May 2026 – Sept. 2026

- Built a real-time GPU-shader renderer that integrates null geodesics via RK4 per-pixel in Kerr-Schild coordinates, with an ImGui interface for live-adjustable black hole mass and spin.
- Modeled physically-based accretion disc colouring and gravitational redshift using Planck's law and CIE colour-matching functions, rendered against a real starfield from 119,625 stars in the HYG stellar database.

EXTRACURRICULARS

Guelph Student Investment Council (GSIC)

Sept. 2025 – Apr. 2026

Analyst

- Analyze public companies and evaluate financial statements, business models, and industry trends to prepare investment theses for council review.

TECHNICAL SKILLS

Languages: C, Python, Java, JavaScript, TypeScript, Dart, GLSL, SQL, HTML/CSS

Frameworks & Libraries: React, Svelte, Vue.js, Django, Django REST Framework, Spring Boot, FastAPI, Flutter, OpenGL, GLFW, ImGui

Data & ML: PostgreSQL, Firebase Firestore, scikit-learn, k-nearest neighbours

Developer Tools: Git, GitHub, VS Code, Docker, Render, Firebase

Practices: REST APIs, Agile Development, Pair Programming, Technical Documentation